

Hypothesis of 3-Brane-Higher-Dimensional Resonant Coupling

Document Note

This paper presents theoretical deductions of a thought experiment. It adopts the conceptual framework of string-theory brane cosmology and gravitational waves. It is **not** an experimentally verified mainstream physical theory. It serves only for logical reasoning to explain phenomena such as ghost sightings, past-life memories, and divinely-revealed information.

Abstract

This thought-experiment model is grounded in brane-cosmology postulates. Gravitational waves act as the sole data-exchange channel between the three-dimensional brane universe and higher-dimensional bulk space. Massive faint gravitational effects generated by neural activity in living brains superimpose to form composite multi-frequency gravitational-wave resonant configurations, which perturb the brane interface and write information into higher-dimensional space.

Resonant configurations are not permanently preserved. During an organism's lifetime, continuous energy input from the brain sustains and reinforces resonant structures to counteract entropy increase. Upon somatic death, energy input ceases; the second law of thermodynamics becomes dominant. Resonant configurations gradually diffuse over time, eventually dissolving completely and merging back into the higher-dimensional background field.

The survival duration of a resonant configuration is positively correlated with its initial strength and phase-locking degree. Stronger obsessions and higher mental-energy density during life produce more stable resonant structures with longer survival times; weaker obsessions lead to faster dissipation. Most extremely ancient resonant configurations have fully dissipated via entropy increase and can no longer produce resonant interactions. Perception of higher-dimensional signals in reality is governed by three factors simultaneously: whether the resonant configuration still exists and its remaining coherence; whether energy-gravitational noise in the three-dimensional environment overwhelms the signal; and whether the recipient's neural-gravitational multi-frequency set matches the target configuration.

Valid trans-brane interaction requires complete matching of the full multi-frequency set; coincidence of only a few frequencies yields no notable effect. Resonant configurations project onto the 3-brane via gravitational waves and generally suffer distance-dependent attenuation. A small number of specially sensitive individuals can modulate their own resonant patterns to raise matching probability. High-noise

environments such as cities and power plants elevate background interference and reduce readout success rates. Low-anthropogenic-disturbance locations such as deep mountains and plateaus facilitate trans-brane resonance. Extreme high-energy events such as nuclear detonations do not directly destroy resonant configurations, yet they cause large-scale signal masking and accelerate entropic dissipation of local configurations. Animals also generate higher-dimensional resonances, though these are weaker and dissipate faster, with multi-frequency coding isolation between species.

Somatic death terminates the maintenance and reinforcement of resonant configurations, initiating their entropic-dissipation process. If a newborn's innate brain-gravitational multi-frequency set matches a not-yet-dissipated resonant configuration, memory fragments may be copied, creating the illusion of reincarnation. Sensitive individuals receiving information from persisting ancient configurations correspond to phenomena of divinely-inspired epics. Experimental verification falls into two phases. At present, neural-control and chronological-statistical studies can be conducted on subjects with exceptional physiological sensitivity. Future work requires high-precision broadband gravitational-wave detectors to perform matching-detection and dissipation-rate measurements in low-noise environments. The full model obeys the speed-of-light limit, causality, and light-cone rules of the three-dimensional universe.

Preliminary Foundational Assumptions

The following are necessary premises for this model; none represent empirically-validated conclusions within modern physics:

1. **Cosmic structure:** Our observable reality constitutes a three-dimensional brane universe embedded within an external higher-dimensional bulk space. Ordinary matter, electromagnetic waves, and photons are confined to the 3-brane. Gravity and gravitational waves penetrate the brane and serve as the sole cross-dimensional information carrier.
2. **Signal generation:** Neural activity in biological brains produces abundant faint gravitational effects, which superimpose to form stable composite multi-frequency gravitational-wave resonant configurations. Muscles and bones cannot generate valid configurations due to insufficient energy density.
3. **Writing and maintenance:** Gravitational-wave perturbations of the 3-brane interface imprint composite multi-frequency resonant configurations into the higher-dimensional bulk space. During an organism's lifetime, continuous energy input from the brain sustains coherence of resonant configurations and even strengthens phase-locking to counteract entropy increase.

4. **Thermodynamic law:** The higher-dimensional bulk space follows the second law of thermodynamics; entropy spontaneously and monotonically increases within isolated systems. After energy input is cut off, ordered composite multi-frequency resonant configurations spontaneously decohere, gradually diffuse, and ultimately dissolve completely, merging into the uniform higher-dimensional background field with their informational structure eliminated.
5. **Survival-duration rule:** Higher initial strength and phase-synchronization strengthen a configuration's resistance to entropy increase and extend its total dissipation time. The intensity of obsessions during life constitutes the core factor determining initial coherence.
6. **Resonant interaction:** Cross-dimensional interaction demands complete matching of the entire multi-frequency ensemble. Overlap of only one or partial frequencies cannot produce conspicuous phenomena. Upon successful matching, higher-dimensional signals act backward on the 3-brane and modulate human cerebral neuroelectrical activity.
7. **Interference mechanism:** Anthropogenic electromagnetic and industrial noise within the three-dimensional world do not directly destroy higher-dimensional resonant configurations, but generate powerful broadband signals that produce local signal masking. Extreme high-energy events such as nuclear detonations, in addition to signal masking, violently perturb the local higher-dimensional field and accelerate entropic dissipation of regional resonant configurations.
8. **Projection and attenuation:** Higher-dimensional resonant configurations exert gravitational-wave-mediated projection onto the 3-brane, generally subject to geographical distance attenuation. Certain sensitive individuals can actively tune their own multi-frequency resonant sets to improve matching efficiency.
9. **Species-level isolation:** Brains of different species generate independent multi-frequency-resonance-coding systems. Resonant matching across species is extremely difficult.
10. **Causal constraints:** All cross-dimensional information interactions must comply with the speed-of-light limit, causality principle, and light-cone rules intrinsic to the three-dimensional brane universe.

I. Core Axioms of the Model

1. Our world is a three-dimensional brane universe surrounded by higher-dimensional bulk space. Ordinary matter, electromagnetic waves, and

photons are trapped inside the brane and cannot directly enter higher-dimensional space. Gravitational oscillations and gravitational-wave radiation traverse the brane boundary and act as the exclusive read-write medium bridging the 3-brane and bulk space.

2. **Writing-and-maintenance mechanism for brane vibrations driven by gravitational waves:** Every material motion and every neural event generates extremely faint gravitational effects. Numerous faint gravitational perturbations from an individual superimpose to form a unique composite multi-frequency gravitational-wave resonant configuration. Resonance is not a single frequency but a combined ensemble of multiple frequencies, analogous to broadband modulation of electromagnetic signals. Vibration at an isolated frequency scarcely yields meaningful interaction; effective resonance occurs only when the full multi-frequency set matches.

Mental activities, obsessions, and memories of all living beings on the 3-brane translate into corresponding gravitational-wave oscillations that perturb the brane-bulk interface and write information into higher-dimensional bulk space. While alive, continuous energy output from the brain sustains and reinforces phase synchronization of this resonant configuration, actively counteracting diffusive effects of entropy increase. Stronger obsessions and higher mental concentration produce more pronounced reinforcement of resonant configurations.

3. **Resonance strength determined by energy density:** Higher energy density of biological neural activity amplifies brane-oscillation amplitude and the intensity of corresponding higher-dimensional resonant fields. Human tissues vary drastically in energy consumption. Muscles and bones consume little metabolic energy and almost never produce valid composite gravitational-wave resonant configurations. The brain is the human organ with the highest energy density. Sustained neuroelectrical activity generates complex superimposed gravitational-wave vibrations that write coherent composite multi-frequency resonant configurations into higher-dimensional space. While alive, consciousness, memory, and mentality reside simultaneously within cerebral neural activity on the 3-brane and within resonant configurations in higher-dimensional space; the two exist symbiotically.
4. **Dual effects of obsession:** The strength of subjective conviction determines two aspects: on one hand, the initial coherence and signal intensity of resonant configurations written into higher-dimensional space; on the other hand, the efficiency of in-life configuration reinforcement, indirectly setting total

anti-entropy survival time after death. Stronger obsessions produce higher phase synchronization within multi-frequency waveforms and more stable structures with longer post-mortem survival times. Weaker obsessions yield faint written signals that dissipate rapidly after death.

5. **Rules for trans-brane readout via resonance:** Effective trans-brane resonance occurs only when the composite multi-frequency gravitational-wave oscillation pattern of a three-dimensional living being sufficiently matches the full frequency ensemble of a not-yet-dissipated higher-dimensional resonant configuration. Mere overlap of a small subset of frequencies cannot trigger substantial interaction. After resonance is established, data stored in higher dimensions acts backward upon the 3-brane to perturb human neuroelectrical activity. This may evoke hallucinations, import memory fragments, or further reinforce the subject's own obsessions.

Corollary: If a human can voluntarily tune composite multi-frequency gravitational-wave oscillations generated by their brain to match more persisting higher-dimensional configurations, they could in principle retrieve more data from higher dimensions. With sufficiently high sensitivity, an individual is released from strict geographical-projection-distance constraints and may read resonant-configuration data originating from different locations and epochs.

6. **Rule of entropic dissipation after death:** Upon somatic death, cerebral neuroelectrical activity ceases. No new gravitational-wave oscillations are produced, and energy input together with phase reinforcement for higher-dimensional resonant configurations terminates. The second law of thermodynamics becomes dominant. Spacetime fluctuations in the higher-dimensional background continuously disrupt phase synchronization of resonant configurations. Structures gradually diffuse starting from their margins; coherence decays monotonically until full dissolution and complete merging into the higher-dimensional background field, with independent informational configurations eliminated.

Core conclusion: No permanently preserved higher-dimensional information backups exist. All resonant configurations gradually dissipate via entropy increase following death, though survival durations differ. Configurations associated with extremely intense obsessions may persist for longer periods, while those of ordinary individuals often fully diffuse within several human generations. Resonant configurations corresponding to extremely ancient historical events have in most cases completely dissipated and can no longer produce resonant interactions.

7. **Influence of environmental energy density on resonant-configuration readout**

- Higher environmental energy density generates abundant chaotic superimposed gravitational perturbations across the 3-brane, elevating the brane-vibration noise floor, masking target signals and lowering matching success probability. Meanwhile, intense noise accelerates phase diffusion of higher-dimensional resonant configurations and shortens their effective survival time.
- Industrial cities, power-generation facilities and large-scale electrical infrastructure produce complex gravitational-energy disturbances and high brane-vibration noise. Resonant configurations left by deceased beings are difficult for three-dimensional life-forms to detect and dissipate faster.
- Remote mountainous and high-plateau regions with minimal human presence and negligible industrial-energy input exhibit weak background brane-vibration noise. Conditions for trans-brane resonance are more readily satisfied; entropy-driven dissipation of resonant configurations slows, extending their effective survival duration.
- Extreme high-energy events (e.g. nuclear-weapon detonations): On one hand, extremely-intense broadband gravitational-wave oscillations propagate into higher-dimensional space and create local signal masking that obscures ordinary resonant configurations. On the other hand, violent energy shocks strongly perturb the local higher-dimensional field and drastically accelerate entropic dissipation of all resonant configurations within the affected region, bringing about their premature disintegration.

8. **Animal-generated higher-dimensional resonance and species-specific frequency isolation:** Animal brains also produce superimposed gravitational-wave oscillations from neural activity and write resonant configurations into higher-dimensional space. Nevertheless, animal brains are smaller with less-complex neural circuits. Neural activity delivers lower energy density and coherence, yielding low-amplitude gravitational-wave resonance and poor anti-entropy capacity. Their configurations dissipate far more rapidly after death compared with humans. Each species possesses its native multi-frequency-resonance ensemble, analogous to linguistic barriers. Gravitational-wave oscillations from animal brains preferentially respond to conspecific higher-dimensional resonances, while human cerebral gravitational-wave signatures show greater sensitivity toward human-origin

resonant-frequency sets. Humans rarely perceive animal-related resonances and animals seldom detect human-related resonances, comparable to human inability to interpret animal vocalizations.

9. **Distance-attenuation principle: projection effect of higher-dimensional configurations onto the 3-brane:** The causal influence exerted by higher-dimensional resonant configurations upon the three-dimensional world manifests as projection of higher-dimensional gravitational-wave vibrations onto the 3-brane. Projection-signal intensity peaks near the original location where the configuration was generated; increasing geographical distance distorts and attenuates projected signals. Ordinarily, trans-brane resonant readout occurs more readily at short range. Exceptionally, subjects with powerful voluntary tuning capability and extreme sensitivity for composite gravitational-wave resonance can transcend geographical-projection-distance limits and retrieve data from resonant configurations stored in higher-dimensional space at remote locations.
10. **Coexistence rule for multiple resonant configurations:** Higher-dimensional bulk space can host multiple independent resonant configurations simultaneously. In the absence of intense-energy shocks, each configuration evolves and dissipates via entropy increase independently without mutual interference or destruction. External high-energy events produce two categories of effects: broadband strong-signal masking that renders target configurations undistinguishable for receivers; and violent energy perturbation that accelerates entropy-increase rates of all configurations and shortens their overall survival duration.
11. **Triple-mechanism for readout barriers:** Failure to observe target higher-dimensional resonances in reality arises from superposition of three factors:
 - Configuration dissipation: Too much time has elapsed and the target resonant configuration has fully dissolved under entropy increase and no longer exists.
 - Signal masking: Environmental-energy noise elevates the background floor, and target-signal intensity falls below the noise threshold and cannot be identified.
 - Coding mismatch: Large divergence exists between the recipient's neural-gravitational multi-frequency set and the target configuration, preventing full resonance even if the signal persists.

Individuals sharing similar ideologies, cognition, and lifestyles exhibit higher similarity in gravitational-wave multi-frequency sets and achieve resonance more easily. Greater temporal separation brings not only larger coding divergence but also higher probability that target configurations have already dissipated, making corresponding phenomena far harder to observe.

12. **Rules for trans-brane resonant coupling: Trans-brane coupling channels activate only when two conditions are met: the full multi-frequency gravitational-wave oscillation suite generated by a living-human brain falls within the resonance window matching the full eigen-frequency ensemble of a higher-dimensional resonant configuration; and the configuration itself retains sufficient residual coherence.** Mere partial frequency overlap or substantial dissipation of the configuration cannot produce functional interaction. Higher-dimensional resonances do not emit visible-light photons within the 3-brane. Once coupling is established, gravitational-wave-driven brane oscillations backward-modulate intracerebral neuroelectrical signals and internally generate subjective percepts including apparitions and auditory hallucinations without requiring light to enter the eyes. Successful reception of resonances depends jointly upon environmental noise, projection distance, residual coherence of the configuration, as well as the recipient's neurophysiological constitution and multi-frequency-set characteristics.
13. **Reincarnation as a perceptual illusion:** When a not-yet-fully-dissipated higher-dimensional gravitational-wave resonant configuration lies near a newborn and matches the multi-frequency gravitational-wave-oscillation set generated by the developing infant brain, trans-brane resonance copies memory-and-emotion fragments into the infant's highly-plastic neural circuits. The original higher-dimensional resonant configuration continues its entropic-dissipation process; this is not translocation of an integral soul entity.

Corollary: High-altitude and remote regions with low industrial-energy flux feature low background gravitational noise on the 3-brane. Resonant configurations survive longer and cross-dimensional information-copying events occur more readily, corresponding to folkloric records of reincarnating child-prophets and divinely-revealed epics. For instance, the "divine-inspiration chanting" of Gesar epic bards arises partly from low-noise high-altitude conditions favouring signal matching, and partly because epic-related resonant configurations have been repeatedly resonated, rewritten and reinforced by successive generations of bards across centuries, counteracting entropic dissipation and extending overall survival duration.

II. Full Evolutionary Sequence Described by the Model

Stage 1: Living-organism phase — Construction, maintenance and reinforcement

Human brains continuously produce neuroelectrical oscillations. Supported by the brain's high energy density, persistent superimposed multi-frequency gravitational-wave vibrations perturb the three-dimensional brane and iteratively write and update coherent composite multi-frequency resonant configurations into higher-dimensional bulk space.

During this phase, continuous energy input from the brain actively counteracts diffusive entropy increase. It not only stabilizes resonant configurations but can further strengthen phase-locking through intense obsessions and highly focused mental activity, raising total intensity and structural robustness and laying the foundation for longer post-mortem survival. Muscles, bones and other tissues have low metabolic-energy consumption and barely generate usable gravitational-wave vibrations; hence they cannot produce identifiable resonant configurations written into higher-dimensional space. Higher-dimensional resonant configurations reside outside the 3-brane and emit no electromagnetic radiation detectable in three-dimensional space, so conventional optical and electromagnetic instruments cannot observe them directly.

Living humans whose full-set gravitational-wave-resonance pattern fully matches a persisting higher-dimensional configuration may conversely experience amplified personal obsessions and receive foreign informational fragments. Such interaction itself supplies minor supplementary energy to the original configuration and slightly slows its entropic dissipation. A minority of individuals capable of voluntarily tuning their own resonant patterns can overcome geographical limits and retrieve data from persisting higher-dimensional configurations originating from distant locations and different time periods.

Stage 2: Somatic death — Onset of entropic dissipation

Cerebral bioelectrical activity terminates. Generation of new gravitational-wave oscillations ceases, halting writing-updates into higher-dimensional storage and cutting off energy maintenance and phase reinforcement for resonant configurations. Entropy-increase effects begin to dominate configuration evolution.

In the immediate post-death period, overall structure and coherence of resonant configurations remain intact, fully preserving memory fragments, emotional signatures and the complete eigen-oscillation multi-frequency ensemble belonging to the deceased subject. At this stage, thought-patterns resemble those of contemporaneous

living humans, yielding high multi-frequency-set-matching probability and raising the likelihood of trans-brane-resonant read-out interactions with suitably-matched living persons.

The environmental gravitational-noise floor directly determines observability of a given configuration and also influences its dissipation rate. High-noise urban environments mask signals and accelerate configuration dissipation. Remote mountains and plateaus with minimal disturbance deliver higher observation probability and longer survival durations. Higher-dimensional resonant configurations project onto the 3-brane via gravitational waves; projection clarity diminishes with increasing distance. Highly-sensitive subjects capable of re-tuning their full multi-frequency resonant-ensemble can capture corresponding signals even at great or trans-regional distances.

Stage 3: Long-term temporal evolution — Gradual diffusion until complete dissolution

With energy input removed, higher-dimensional background-spacetime fluctuations continuously perturb resonant configurations. Phase-synchronization declines progressively, structures diffuse starting from their margins, and coherence decays monotonically. This process is irreversible. Eventually the entire resonant configuration dissolves completely and merges into the higher-dimensional background field, eliminating its independent informational structure.

Core factors governing dissipation rate:

1. **Initial strength and phase-locking degree:** Stronger obsessions during life and higher initial configuration coherence improve resistance to entropy-increase and extend total time required for full dissipation.
2. **Environmental-perturbation level:** Stronger three-dimensional-side energy noise translates into more perturbations penetrating into higher dimensions, accelerating phase diffusion and shortening configuration survival time.
3. **Reinforcement via repeated resonance:** If configurations are repeatedly read and resonated by later-era humans, indirect supplementary energy is obtained, slowing entropy-increase and greatly extending survival duration.

Extreme high-energy shocks (e.g. nuclear detonations) produce two-fold impacts: instantaneous strong-signal masking of all target configurations leading to temporary unobservability; and violent perturbation of the local higher-dimensional field that drastically accelerates entropy-increase rates of regional configurations and brings about their premature dissolution. No permanent higher-dimensional information

records exist. Resonant configurations belonging to most ordinary individuals gradually diffuse and vanish within decades to centuries after death. Only configurations associated with extremely intense obsessions or repeatedly reinforced by later-era resonance may survive for thousands of years. Resonant configurations corresponding to sufficiently ancient historical events have in most cases fully dissipated and can no longer be observed.

III. Explanation of Observed Phenomena Under This Model

1. **Ghost-sighting phenomena:** Not-yet-dissipated composite multi-frequency gravitational-wave resonant configurations in higher-dimensional space project onto the 3-brane through gravitational-wave oscillations. When encountering living humans whose full-ensemble gravitational-wave-vibration set falls within the resonance window, trans-brane resonant-coupling activates and directly modulates the subject's visual and auditory cortices, generating subjective percepts including human-shaped apparitions and whispered voices. Mere partial frequency overlap or substantial dissipation of the configuration cannot trigger valid phenomena.
 - Conventional cameras record nothing: no visible-light photons are emitted within the three-dimensional brane, so optical apparatus yields no trace.
 - Inter-subject variability: Only persons whose full cerebral-gravitational-electrical multi-frequency-vibration-set satisfies resonance conditions perceive these manifestations; co-present observers remain unaffected. Percept quality correlates with the recipient's neurophysiology and native full-ensemble gravitational-resonance set.
 - Distance dependence: Projection clarity and detection probability decrease with geographical separation. Exceptionally sensitive subjects with voluntary tuning-capability for their full multi-frequency resonant-ensemble can detect higher-dimensional signals originating from far-off locations.
 - Temporal effects: Resonant configurations of long-deceased individuals exhibit higher degrees of dissipation, lowering observability probability. Recently-deceased subjects are more likely to be witnessed since configuration coherence remains maximal.
 - Geographical variability: Remote mountainous and plateau regions with

low anthropogenic gravitational-noise support longer-lived configurations and higher read-out probability. Observations are rare within cities, power-plant vicinities, and nuclear-detonation sites subject to high noise and intense perturbation.

2. **Dual effects of obsession:** Powerful pre-mortem obsessions and emotions strengthen phase-locking within full multi-frequency gravitational-wave waveforms written to higher-dimensional space, raising initial intensity and extending post-mortem survival duration. Weak obsessions produce faint written signals that dissipate rapidly. Living humans undergoing complete-ensemble resonance with persisting higher-dimensional resonant configurations may conversely experience amplified personal obsessions and emotional states; such interaction also slightly slows entropic dissipation of target configurations.
3. **Weak resonant signatures from animals and species-specific frequency barriers:** Animal brains are small with limited neural complexity. Insufficient energy density in neural activity yields faint superimposed gravitational-wave-oscillations and poor anti-entropy capacity. Their resonant configurations dissipate far faster after death compared with humans. Large-scale divergence exists between human and animal full-ensemble multi-frequency resonant-configurations, erecting linguistic-like barriers. Human cerebral gravitational-wave responses preferentially detect human-origin higher-dimensional resonances, while animals primarily respond to conspecific signatures. Cross-species perception rarely occurs.
4. **Past-life memories, reincarnating-child accounts, and divinely-inspired-epic phenomena:** High-plateau environments such as the Tibetan Plateau exhibit minimal anthropogenic industrial-energy flux and low gravitational background perturbation within the three-dimensional brane. Resonant configurations survive longer and become more accessible for matched read-out. When the full multi-frequency gravitational-wave-oscillation-set generated by a nearby developing-infant brain matches a not-yet-dissipated configuration, trans-brane resonance copies memory-and-emotion fragments.
 - This yields informational copies rather than translocation of an integral soul entity; the original configuration continues entropic dissipation.
 - Memories are mostly fragmentary partly because target configurations have partially diffused and deliver incomplete signals, and partly

because the infant's maturing brain reconfigures its neural-gravitational multi-frequency-set and loses matching conditions over time, leading to gradual forgetting.

- The “divine-inspiration chanting” of Gesar epic bards benefits from low-noise plateau-environment matching conditions. Furthermore, epic-related resonant configurations have been repeatedly read and resonantly rewritten across centuries by successive bards, receiving continuous supplementary energy that counteracts entropy-increase and greatly extends survival duration.
- In principle, one single higher-dimensional resonant configuration can sequentially couple to multiple infants or sensitive individuals, producing similar memory fragments across distinct recipients.
- Resonant configurations for most extremely ancient events have fully dissipated; hence retrievable ancient information is extremely scarce and mostly fragmentary.

5. **Locations predisposed to paranormal-phenomenon reports:** Local gravitational micro-perturbations and dark-energy background fluctuations superimposed upon low anthropogenic gravitational-energy disturbance suppress the brane-vibration noise-floor locally, meanwhile slowing entropy-increase rates of higher-dimensional resonant configurations, extending survival durations and raising matching-and-decoding success rates. This creates persistently “haunted” sites. The effect arises not from greater numbers of configurations at that location but from improved read-out conditions due to lower noise and slower dissipation.

6. **Impacts of extreme high-energy events:** Energy releases such as nuclear detonations generate large-amplitude far-reaching gravitational-wave-vibrations that penetrate into higher-dimensional space, producing two-fold effects:

- Signal masking: Broadband intense signals obscure other regional resonant-configurations, rendering them temporarily undiscernible for three-dimensional-side receivers.
- Accelerated dissipation: Violent energy shocks strongly perturb the local higher-dimensional field and accelerate phase diffusion of all regional resonant-configurations, bringing about their premature entropic dissolution.

Over time, the intense signals generated by the detonation itself also gradually dissipate via entropy-increase and the local field returns toward equilibrium. However, previously-accelerated-dissipated original configurations cannot be restored.

7. **Coexistence of multiple configurations:** Higher-dimensional bulk-space can host many independent composite multi-frequency gravitational-wave resonant configurations simultaneously. Each evolves and dissipates independently without mutual destructive superposition. Provided no intense-energy shocks occur, numerous resonant-configurations can coexist within the same higher-dimensional spatial region.

IV. Thought-Experiment for Resonance Detection (Validation Protocol)

Experimental testing of this model imposes stringent hardware requirements. An ultra-precision broadband gravitational-wave detector capable of capturing exceedingly faint composite multi-frequency gravitational-wave variations originating from human biological activity is prerequisite for investigations targeting higher-dimensional resonance. While suitable instrumentation remains undeveloped, only preliminary human-subject studies can proceed.

The complete experiment divides into two phases:

Phase One: Preliminary human-subject investigations (feasible under present-day-technology constraints)

In the absence of qualified precision broadband gravitational-wave detectors, direct acquisition of higher-dimensional signals cannot be performed. Comparative-study protocols may focus first upon self-reported sensitive subjects claiming perception of higher-dimensional-derived information:

1. Acquire electroencephalographic and neural-activity datasets from these subjects alongside matched control-subject groups.
2. Compare cerebral-activation patterns between groups during episodes when sensitive subjects report reception of "higher-dimensional information".
3. Characterize neurophysiological correlates of sensitive-subject populations to guide future-instrument-development.
4. Chronological-gradient statistics: Control environmental and cultural variables, count perception success rates for targets from different post-mortem time-points, fit dissipation-rate curves of resonant configurations and verify entropy-driven temporal-decay laws.

5. **Environmental-control experiments:** Repeat tests across environments with varying noise levels to verify environmental-noise impacts upon perception success rates.

This phase relies entirely upon subjective human reports for signal discrimination. Strong subject-dependent variability limits strict scientific reproducibility; this work constitutes purely exploratory preliminary research.

Phase Two: Instrument-based physical detection (to be executed after future development of precision broadband gravitational-wave detectors)

1. **Site-selection:** Select remote underground / wilderness locations with minimal anthropogenic energy-related and gravitational-noise sources to minimize environmental signal-interference.
2. **Baseline-signal acquisition:** Record the complete composite multi-frequency gravitational-wave-configuration produced by a living-human test-subject at the experimental site.
3. **Simulated-configuration reception test:** Operate the detector to reconstruct and simulate the recorded full-set human-origin multi-frequency gravitational-wave-configuration as a matching template. Employ this template in attempts to match and receive corresponding resonant signals originating from higher-dimensional space.
4. **Dissipation-rate-measurement experiment:** Artificially inject known standard multi-frequency resonant signals, monitor long-term changes in coherence and intensity attenuation, directly measure entropy-driven dissipation rates of gravitational-wave resonances, and verify survival-duration differences under varying initial-intensity conditions.
5. **Interpretation-of-results:** Stable detection of extra resonant-signals correlating with the template and exhibiting time-dependent intensity decay matching theoretical predictions would furnish tentative supporting evidence for the hypothesis. Failure to detect matching signals cannot validate the hypothesis.

Supplementary-control approach: Iterate testing using diverse template-configurations. Compare signal-detection rates across different templates to test the core postulate requiring full-ensemble multi-frequency matching.

V. Premises and Intrinsic Limitations of the Model

Additional postulates required (outside established standard physics)

1. The energy-density of human-cerebral neural-activity determines amplitude of superimposed human-origin gravitational-wave-oscillations. High-energy cerebral output constitutes a necessary condition for writing valid composite multi-frequency resonant configurations into higher-dimensional space.
2. Enormous populations of faint gravitational-effects generated by biological-life-activity can superimpose to stabilize composite multi-frequency gravitational-wave resonant configurations. Isolated-frequency oscillation cannot sustain meaningful resonant-interaction. Gravitational-wave-vibrations traverse the three-dimensional brane boundary and implement read-write data-transfer between the 3-brane universe and higher-dimensional bulk space.
3. While alive, the brain can sustain and reinforce higher-dimensional resonant configurations via continuous energy input to counteract entropy-increase. After death, energy input terminates. Higher-dimensional resonant configurations follow the second law of thermodynamics, undergoing decoherence and diffusion driven by entropy-increase until full dissolution with elimination of informational structure.
4. Survival duration of resonant configurations is positively correlated with initial coherence; obsession intensity serves as the core factor determining initial coherence. Repeated post-humous resonance-readout by later-era subjects indirectly supplies energy, slows entropy-increase and extends survival duration.
5. Electromagnetic and high-energy events within the three-dimensional brane produce superimposed gravitational-wave-oscillations that cross the brane boundary and generate signatures inside higher-dimensional space. High-energy-events produce signal-masking and simultaneously accelerate entropic dissipation of local resonant configurations.
6. Higher-dimensional resonant-configurations manifest upon the three-dimensional brane via gravitational-wave-mediated projection subject to distance-dependent attenuation. Exceptionally sensitive individuals capable of voluntary re-tuning of their full multi-frequency resonant-ensemble can improve matching probability.
7. Higher-dimensional bulk-space permits coexistence of multiple resonant-configurations, each evolving and dissipating independently without mutual destruction.
8. Distinct biological-species possess independent full-ensemble multi-frequency

resonant-coding-systems; resonant matching across species is extremely difficult.

9. Gravitational-wave-driven higher-dimensional resonant-configurations can encode memory, personality and emotional information. Read-out demands full-ensemble-matching with the recipient's native multi-frequency-set and is also constrained by residual coherence of target configurations.
10. Read-out barriers originate from three sources: ① target-configurations have dissipated via entropy-increase; ② signal masking caused by three-dimensional-side environmental-noise; ③ excessive multi-frequency-coding divergence between recipient and target-configuration preventing resonant decoding.

Outstanding unresolved gaps within the model

1. Quantitative projection-attenuation relationship: Mathematical laws governing how higher-dimensional gravitational-wave-projection weakens with geographical distance remain unspecified.
2. Gravitational-coupling-channel mechanism: How extremely faint gravitational-effects stemming from biological-neural-activity superimpose into stable composite multi-frequency resonant-configurations, and the underlying coupling-physics enabling gravitational-waves to implement brane-crossing data-reading-and-writing, require further elaboration.
3. Quantitative dissipation-rate model: The entropy-driven dissipation timescale for biological-origin gravitational-wave resonances, plus quantitative impacts of obsession intensity and environmental noise upon dissipation rates, remain undescribed.
4. Quantification of resonance-reinforcement effects: The magnitude of survival-duration extension produced by repeated readout-resonance, and whether sustained resonance can achieve near-permanent maintenance of configurations, remain unclear.
5. Physiological-basis for sensitive subjects: Special cerebral-neural characteristics enabling certain persons to voluntarily re-configure their full multi-frequency gravitational-wave-resonant-ensemble and achieve long-distance read-out await explanatory development.
6. Animal-resonance boundaries: Criteria delineating species-specific full-ensemble multi-frequency resonant-coding boundaries, and identification

of which animal brains possess sufficient energy-density to generate observable higher-dimensional resonances, remain undefined.

7. Physiological pathways for voluntary individual re-configuration of full-ensemble multi-frequency gravitational-wave-oscillations: Biological mechanisms allowing organisms to modify their superimposed gravitational-wave multi-frequency-pattern are not described.
8. Partial-matching-scenarios: Subjective phenomenology triggered when recipients overlap only a frequency subset of a stored-configuration lacks precise definition.
9. Quantitative impacts of extreme energy: Magnitude of dissipation-rate elevation for local configurations caused by events such as nuclear detonations, plus the dissipation timescale of their own generated signals, have no quantitative description.

VI. Summary

This hypothesis integrates ghost sightings, paranormal encounters, past-life memories, divinely-inspired epics and related phenomena into one internally-consistent conceptual framework. Ghost-like manifestations fundamentally correspond to composite multi-frequency resonant informational configurations written into higher-dimensional space via post-mortem cerebral gravitational-wave-driven brane-interface vibrations. Data-reading-and-writing across the boundary separating three-dimensional-brane-universe and higher-dimensional-bulk-space are mediated by brane-penetrating gravitational-waves. Resonant configurations are not permanent: during life, continuous energy input from the brain sustains and reinforces resonant structures to counteract entropy-increase. Upon somatic death energy input ceases; entropy-increase becomes dominant and configurations gradually diffuse and ultimately dissolve completely, merging back into the higher-dimensional background field.

Survival duration of resonant configurations is determined by initial strength and phase-locking degree. Stronger obsessions during life produce more stable structures with longer survival times. Repeated resonant read-out by later-era subjects indirectly supplies energy and slows dissipation. Resonant configurations corresponding to ancient events have mostly dissipated via entropy-increase and are therefore difficult to observe. Successful read-out of higher-dimensional signals depends jointly upon three factors: whether configurations persist, environmental-noise levels, and coding-matching degree on the receiving side. Resonance requires complete matching of the full multi-frequency-ensemble; isolated or partial-frequency-overlap exerts

negligible influence.

Higher-dimensional resonances manifest within the three-dimensional-world as gravitational-wave-projection-effects; ordinarily signal strength declines with increasing distance. A small population of extremely-sensitive subjects can tune their own resonant patterns to raise matching probability. Cities and power-plant-sites constitute high-noise environments that elevate background interference and simultaneously accelerate configuration dissipation. Remote mountain-and-plateau-regions with minimal anthropogenic disturbance support longer-lived configurations and facilitate trans-brane-resonance interactions. Extreme high-energy events such as nuclear detonations produce signal masking and accelerate dissipation of local configurations. Animals also generate higher-dimensional resonances, yet these are weaker and dissipate faster, with species-specific coding isolation.

Obsession-intensity determines initial configuration strength and survival duration. Living-human subjects can resonate with persisting higher-dimensional configurations, experiencing reciprocal amplification of personal obsessions while slowing dissipation of target configurations. Trans-brane-resonant-coupling can directly perturb human neuroelectrical-signals to produce hallucinatory percepts or copy memory-fragments into neonatal brains, generating subjective illusions of reincarnation and divinely-imparted-knowledge. The complete model respects speed-of-light constraints, causality, and light-cone geometry valid within the three-dimensional-brane-universe. Cross-dimensional-information-read-write-operations rely upon gravitational-wave-driven brane-interface-vibrations and occur exclusively when two prerequisites are satisfied: configurations persist and full-ensemble multi-frequency-resonant-set-matching is achieved.